Weekly Report: [W11]

General Information

- **Group ID:** 52
- **Group Name:** Group 52
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-08-16 – 2026-08-22
- **Github Repository:** https://github.com/ndmhuy/SuperMarioGame.git

Pre-flight (per `docs/REPORT_RULES.md` §2.1): `git fetch --all` run before any line below was written. `git rev-list --left-right --count dev...origin/dev` → `0 0` — local `dev` is exactly in sync with `origin/dev`, no staleness. `git log --since= '2026-08-16' --until='2026-08-22 23:59:59' --format='%h' | wc -l` → **100 commits on `dev`**, matching the count in the orchestrating sweep plan's §4.3 exactly (that plan document is a local planning artifact outside this branch's tracked tree, not part of this deliverable, so it is not linked here). `git log --all` over the same window returns 159 commits; the extra 59 are `FubuGold`'s unrelated `A/mapgen-gan-plan` history and other already-merged branch tips reachable a second time through their own refs — not additional week content (see "Branches" below for the one real exception).

Tasks Completed This Week

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities)

98 of the 100 commits on `dev` this week are authored by Member A. Work moved through eight task branches, each merged back into `dev` the same week (`git log --merges --since=... --until=...` lists seven merge commits: `43ff0ca`, `dcc1214`, `0beae5e`, `d22669e`, `75ba6f6`, `09b6e87`, `cca1a59`).

Branch: `A/fix/audit-remediation` / `A/fix/member-b-findings` (audit and remediation)

- **Task 1: Full-project code audit — 37 findings**: Authored `docs/issues/code_audit_2026-08-18.md` (commit `4d0fa31`), a two-domain review recording 7 critical, 14 high, and 16 medium findings, plus `docs/issues/member_a_fix_plan.md` grouping Member A's share into 12 work packages (WP1–WP12).
- **Task 2: WP1–WP4 — coin tiles, dangling player, decorator UAF, CTest** (`d821e0c`): fixed a coin-tile double-award bug in level data handling, a dangling `Player*` left behind after a form change, a use-after-free in the Star/Mega decorator chain, and re-registered the missing CTest cases.
- **Task 3: WP5–WP8 — rewind, head-butt, jump feel, decorators** (`7fd533c`) and a tile-render culling / dead-field cleanup pass (`3072e43`) in `PlayingState.cpp`.
- **Task 4: WP9–WP12 — cast-based dispatch replaced with virtuals**: refactored enemy/physics collision dispatch from `dynamic_cast` chains to virtual calls (`e2479e5`), collapsed sprite rendering to one renderer and decoupled the physics AABB from drawing (`97a6f21`), and extracted the dev ImGui panels into `DevPanel` so `render()` no longer mutates state (`9b5f941`). Merged as `09b6e87` — "all 12 work packages."
- **Task 5: Camera clamping and gameplay completion (C-1..C-4, G-1..G-4, B-2)** (`cd4724c`, merged `0beae5e`): clamped the camera to level bounds and closed four gameplay-completion findings from the audit.
- **Task 6: Remaining Member B audit findings (B-3,5,6,7,8,10,11,12,14)** (`72c53d6`, merged `d22669e`): closed the audit items filed against Member B's domain that Member A's remediation plan had picked up.
- **Task 7: Campaign population and asset sync** (`70d1ce0`, merged `75ba6f6`): populated the 3-world campaign, gave question blocks real contents, and synchronized the stale `build/assets` copy with `assets/`.

Branch: `A/tier1-game-loop` (Tier 1 — the game-loop front end)

- **Task 8: State stack + full game loop**: implemented `GameStateManager` driving the whole state stack (`91ab4d4`), then `PauseState`, `GameOverState`, `VictoryState`, `CharacterSelectState` and `OptionsState` (`0b908aa`).
- **Task 9: Front-end regression coverage**: added `verify_frontend_states.cpp` (396 new lines) and 263 new lines in `verify_regressions.cpp`, and fixed a member-initializer reorder warning in `MovingPlatform.cpp` (`cd41764`).
- **Task 10: Save Game and death counter, real sprites** (`9f2c233`, `0dcbb84`): wired save-on-pause in `PauseState.cpp`, and replaced placeholder flag/power-up art with the real atlas frames in `Flagpole.cpp`.

Branch: `A/tier2-bosses` (Tier 2 — boss fights)

- **Task 11: Boss foundation**: added `Boss.hpp` and `Projectile.hpp` as the shared base both bosses need (`d3e8b55`).
- **Task 12: Bowser boss fight** (`fc6a6c9`): implemented `Bowser.hpp` / `.cpp` and `BossFireball` , wired into `assets/levels/level_3.json` .
- **Task 13: Boom Boom mid-boss** (`78b2786`): implemented `BoomBoom.hpp` / `.cpp` and registered it in `EntityFactory.cpp` , wired into `assets/levels/level_2.json` .
- **Task 14: Difficulty setting** (`90bbdb1`): added `DifficultyStrategy.hpp` / `.cpp` and consumed it from `Enemy.hpp` .
- **Task 15: World Map state** (`8c7eccf`): added `WorldMapState.cpp` (274 new lines) and `CampaignProgress.cpp` (128 new lines).
- **Task 16: Thwomp state machine and AI overlay** (`91dbccb`): finished item 9.1 — Thwomp's slam/rise cycle and an ImGui AI-state overlay across `ChaseStrategy` , `FlyStrategy` and `HammerThrowStrategy` .

Branch: `A/tier3-systems` (Tier 3 — meta-systems)

- **Task 17: Fireball gate, controls, enemy speeds** (`dd85898`): fixed a fire-mode gate and remapped controls in `InputManager.hpp` , added real fireball art in `Fireball.hpp` .
- **Task 18: Mega Mushroom shape fix** (`c75d9a8`): fixed `IPlayerState.cpp` so the Mega decorator preserves the wrapped form's collision shape instead of resetting it.
- **Task 19: ObjectPool for projectiles** (`6aa5654`): implemented `ObjectPool.hpp` and pooled `Fireball` , `Hammer` , and `BossFireball` allocations.
- **Task 20: entities.json + parallax + water/lava + colourblind + New Game+ + debug console + replay + 2-player versus** (`ad88b46` , `eaeaf46` , `e0bb734` , `2e7f5bb` , `5b8e396` , `4ada509` , `cde613c` , `a64691e`): read entity type names from `entities.json` instead of hard-coded strings; added the parallax `BackgroundRenderer` ; animated water/lava tiles and camera lookahead in `Camera.hpp` ; added `ColorPalette.hpp` / `.cpp` for colourblind-mode remapping and audio cues in `SoundManager.cpp` ; added `MetaGame.hpp` / `.cpp` for New Game+/daily challenge/ unlockables; added `DebugConsole.hpp` / `.cpp` (322 new lines, Command pattern); added `ReplayRecorder.hpp` ; and extended `PlayingState.cpp` with two-player versus mode (132 new lines).

Branch: `A/complete-game-tiers-1-4` and `A/shadow-mario-ai-multiplayer`

- **Task 21: Shadow Mario, AI opponent, four multiplayer modes** (`b878f38`): fixed three broken two-player foundations first — `PlayingState::handleInput` only ever dispatched press events to Player 1 (P2 had hold-only mappings and could never jump); the minimap toggle and P2's fire key were both bound to `M` ; sixteen enemy AI call sites asked `Game::getPlayer()` (always P1) instead of `getNearestPlayer()`. Then implemented `ShadowMario`, a `Player` subclass driven by a 60 Hz deque of `PlayerFramePacket` replayed at a 3 s delay with a lerp back onto the recorded position — deliberately not built on `ReplayRecorder`, which records state (not input) and decimates to every 6th frame. Rendered through `SpriteColorFilter`, closing audit item B-9.

- **Task 22: CI made real, twelve teammate playtest defects** (`8141120`, `33d5aae`, `4ac8a1c`, `03bf828`, `60cc735`, `fa318b7`, `0c9c173`, `f6fc8bc`): the CI workflow at `.github/workflows/ci.yml` built only 2 of the 13 binaries `ctest` then tried to run, so every run in the project's history — `dev` included — reported red for reasons that had nothing to do with anyone's code (`8141120`); replaced the hand-written binary list with a `ctest_binaries` target derived from test registration. Fixed 4 harnesses that opened a real window on a headless CI runner (`33d5aae`); fixed 28 files using `std::` symbols without including their headers (`4ac8a1c`); fixed `CMakeLists.txt` naming `src/Graphics/HUD.cpp` when the file is `Hud.cpp` (`03bf828`); fixed a can't-jump-while-walking-into-a-wall physics bug (`60cc735`); fixed window/texture cache teardown order on exit (`fa318b7`); read control hints from the actual key bindings in `MenuState` (`0c9c173`); and closed twelve defects from Member B's playtest plus two overflowing menus (`f6fc8bc`). Merged as `cca1a59` — "the first green CI" — and delivered to `main` (`f0a61f4`, PR #13).

***Branch:* `A/fix/windows-crash-and-playtest`**

- **Task 23: The Level 1-3 crash, the buried backdrop, two dead items, and twelve teammate playtest defects** (`4f80862`, `da8ca99`, `ee0343b`, `14f215f`, `cb4f6ea`): see Issue 1 below for the crash. Fixed `BackgroundRenderer` misclassifying every ground decoration layer as sky (`da8ca99`); fixed the HUD world label, Player 2 badge and a void-death rewind lock (`ee0343b`); built a winnable Bowser fight with a stagger window and a lava-pit bridge ending for World 1-3 (`BridgeAxe.cpp`, `Castle.cpp`, `14f215f`); and replaced three hand-written, drifted copies of the entity type list (in `EntityFactory`, `SerializationUtils`, and the level editor's palette) with one `EntityCatalogue` both the parser and the palette read (`cb4f6ea`).
- **Task 24: Test hermeticity** (`045f9b7`): see Issue 4 below.
- **Task 25: Reports and learning record** (`3c0a768`, `64088b4`, `a0395d6`, `fdec9cc`, `6c946f6`): generated the CS202 final report from `reports/report_content.py` via `reports/build_report.py`, a LaTeX/PDF edition under `Report/SuperMarioGame/`, a UML class diagram generated from headers by `tools/gen_class_diagram.py`, and a learning record at `docs/learning/mid-frame-entity-spawn-crash.html` with excerpts extracted by `tools/extract_learning_excerpts.py`.

25125084 - Trần Gia Huy — "FubuGold" (Member B - Input, Sound & Gameplay Systems)

Member B's share of W11 is **2 commits on `dev`**, both authored by the git identity `FubuGold`. This is reported plainly rather than padded — see "Not verified / plan correction" below for how this reconciles with the sweep plan's "5 commits by `FubuGold` since 08-16."

Branch: `B/feature-fix/enemies-behavior`

- **Task 1: Enemy AI tuning across all 11 enemy types** (`d954a66`, 2026-08-16, merged into `dev` via `43ff0ca` on 2026-08-18): tuned `Enemy.cpp`, `KoopaTroopa.cpp`, `KoopaParatroopa.cpp`, `Thwomp.cpp`, `Boo.cpp` and `HammerThrowStrategy.cpp`. Concretely: `Enemy::getBoundingBox` now returns an empty AABB while `isDeadOrDying()`, so `CollisionResolver` no longer resolves against a squished corpse; Goomba squishes right-side-up for 0.5 s then plunges out of the world unflipped, while a flipped/stomped death elsewhere applies 3500 px/s² downward gravity; `KoopaTroopa` gained a `ShellHeld` state (`pickUp()` / `throwShell()` in `KoopaTroopa.cpp` lines 216–240) so touching a shell from the side picks it up and pressing `F` throws it at 550 px/s and 45°; `KoopaParatroopa` gained a 1 s transformation-invincibility window on losing its wings; and `HammerThrowStrategy` applies a random ±10% velocity variance so consecutive hammer throws are not identical.
- This work is the squashed result of nine iterative sessions logged between 2026-08-16 19:40 and 22:47 under `logs/agent_history.log` (each session's own entry reads "no new commits per user directive" — the changes only became a real commit once bundled into `d954a66`).

Branch: `A/fix/windows-crash-and-playtest` (verification only)

- **Task 2: Windows verification of the Level 1-3 crash fix** (`5bc27c1`, 2026-08-22): this commit's diff touches only `logs/agent_history.log` and `saves/config.json` — **no source files.** Its log entry records running `SuperMarioGame.exe` on Windows through the Bowser fire-breath sequence and observing zero crashes, closing the "NOT CONFIRMED ON WINDOWS" gap that Member A's `4f80862` fix (Issue 1 below) had left open at 16:05 the same day (`4f80862` lands at 14:58, `5bc27c1` at 21:57 — see Issue 1). The commit message "fix: fix crash in level 1.3 in window" is more accurately a Windows *re-verification* of an already-committed fix, not an independent fix, and this report corrects that framing rather than repeating it.

AI Usage Declaration

- Used AI to author the full-project code audit (`docs/issues/code_audit_2026-08-18.md`) and triage its 37 findings into Member A's 12-work-package remediation plan.
- Used AI to design and implement `IMovementStrategy` -conformant enemy behavior tuning (Thwomp slam/rise timing, Boo gaze detection, HammerBro throw variance, Koopa shell pickup/throw state machine) on `B/feature-fix/enemies-behavior` .
- Used AI to diagnose the Level 1-3 crash as a mid-frame `std::vector` reallocation invalidating loop iterators (`PlayingState::update` / `PhysicsEngine::update` iterating `m_entities` while a synchronous `EventBus` spawn handler pushed onto the same vector), and to design the deferred-spawn queue that fixes it.
- Used AI to implement the Shadow Mario AI opponent's frame-delayed playback (`ShadowMario` + `PlayerFramePacket` ring buffer) and the four multiplayer modes.
- Used AI to build the CI pipeline fix (deriving the tested-binary list from test registration via a `ctest_binaries` CMake target) and to diagnose the 28-file missing- `<algorithm>` / `<cstdint>` -style include gaps caught only once CI actually built every target.
- Used AI to generate the UML class diagram from `include/**/*.hpp` (`tools/gen_class_diagram.py`) and the learning record excerpt extractor (`tools/extract_learning_excerpts.py`).

Tasks Planned for Next Week

- **Nguyễn Đình Minh Huy**: per `logs/agent_history.log` 's 2026-08-22 session, the outstanding items recorded at the time were: play through the new Bowser stagger fight, P-Switch, and POW block manually rather than only via `ctest` /scripted runs; confirm the bonus/procedural castle endings beyond World 1-3; and tune the stagger numbers (4 fireballs, 3 s window) if they play too easy or too hard. (W12's actual git history shows only 1 commit that week — reported separately in `docs/Group52_12/52.md` , out of scope for this document.)
- **Trần Gia Huy**: continue enemy-behavior tuning and playtest passes per the `B/feature-fix/enemies-behavior` branch pattern; no forward-looking plan for Member B was recorded in `logs/agent_history.log` for this week beyond that.

Issues & Resolutions

Issue 1: Level 1-3 crashes on Windows when Bowser fires his first fireball

- **Problem**: Three separate `EventBus` handlers — `PowerUpRequested`, `EntitySpawnRequested`, and `PlayerShotFireball` — each called `m_entities.push_back(...)` directly from inside a *synchronous* event callback in `PlayingState.cpp`. Those callbacks fire on the call stack of `PlayingState::update()`'s range-`for` over `m_entities` and of `PhysicsEngine::update()`, both of which are mid-iteration over that exact vector at the time. A `push_back` that triggers a reallocation invalidates every iterator and reference the enclosing loop holds — undefined behaviour. On macOS the allocator's spare capacity usually absorbed the reallocation silently; on Windows it did not. World 1-3 is the only level where Bowser fires a projectile every 1.2–2.2 s for the entire boss fight, which is why an earlier workaround of simply deleting Bowser from the level "fixed" the crash.
- **Resolution**: Commit `4f80862` (2026-08-22 14:58, `fix(Core): the 1-3 crash, the buried backdrop, and two items that did nothing`) introduced `m_pendingSpawns`, a staging buffer that every spawn-triggering event now pushes onto instead of `m_entities` directly. A single `flushPendingSpawns()` call drains that buffer into `m_entities` once per frame, after both the `PlayingState::update()` range-`for` and `PhysicsEngine::update()` have finished iterating — so no loop is ever holding an iterator into a vector that gets resized underneath it. The fix was verified on macOS the same day (`ctest` 13/13, scripted run holding Level 1-3 open ~11 s past Bowser's first fireballs) but the crash itself was Windows-only and so remained unconfirmed there until commit `5bc27c1` (2026-08-22 21:57, authored by `FubuGold`), which ran `SuperMarioGame.exe` on Windows through the same Bowser fire-breath sequence and recorded zero crashes — closing the gap the same day it was opened. A self-contained walkthrough of the failure and the fix lives at `docs/learning/mid-frame-entity-spawn-crash.html`.

Issue 2: Two-player mode was three separate broken features, not one

- **Problem**: `PlayingState::handleInput` only ever routed key-press events to Player 1; Player 2's bindings covered hold-based actions (walk, crouch) but no press-based action ever reached P2, so Player 2 could never jump in any two-player mode. Separately, the minimap toggle and Player 2's fire-fireball key were both bound to `M`, so pressing it only ever opened the minimap. Separately again, every enemy AI strategy queried `Game::getPlayer()`, which by definition returns Player 1, so all sixteen call sites that used it ignored Player 2 entirely — enemies would path toward P1 even while standing on top of P2.

- **Resolution**: Commit `b878f38` (`feat(Core, Entities): Shadow Mario, an AI opponent, and four multiplayer modes`) fixed all three as prerequisites before adding new multiplayer content: `handleInput` now dispatches press events per player index; P2's fire binding was moved off `M`; and the sixteen call sites in `ChaseStrategy` and related strategy classes were changed from `Game::getPlayer()` to `Game::getNearestPlayer()`, which selects by proximity rather than by a hard-coded player-1 assumption.

Issue 3: The CI workflow was red for every commit in the project's history for a reason unrelated to code quality

- **Problem**: The GitHub Actions workflow at `.github/workflows/ci.yml` built exactly two binaries — `SuperMarioGame` and `verify_regressions` — via a hand-written CMake target list, then invoked `ctest`, which tried to run thirteen registered test cases. Eleven of them reported `***Not Run` because their executables were never built, and the job failed on every push to every branch including `dev`, regardless of whether the underlying code was correct.
- **Resolution**: Commit `8141120` (`fix(ci): the workflow built 2 of the 13 binaries it then tested`) replaced the manually maintained binary list with a `ctest_binaries` CMake target that `add_verify_test()` attaches every registered case to as it is declared, so the set of binaries CI builds is derived from the same registration call that adds the test — a new test can no longer go unbuilt by omission. Follow-up commit `33d5aae` fixed four of those harnesses opening a real `sf::RenderWindow` on the headless GitHub Actions runner (no display available), and `4ac8a1c` fixed 28 files that used `std::` library symbols without including the headers that declare them — defects that had been invisible until CI actually compiled every target for the first time.

Issue 4: `ctest` runs corrupted the developer's real save data

- **Problem**: `CampaignProgress::reset()` calls through `Serializer::saveDirectory()`, which resolves a save path relative to the process's *working directory* rather than to a fixed location. Existing regression tests worked around this with hand-written backup/restore logic that renamed `saves/progress.json` before and after each run — but that rename was itself relative to the working directory, and the harnesses were invoked from `build/`, so the workaround protected nothing: a live `ctest` run deleted the developer's actual `saves/progress.json`, and a high-score regression case passed or failed depending on what values already happened to be on disk.

- **Resolution**: Commit `045f9b7` (`fix(tests): the suite can no longer read or delete real save data`) added `Serializer::setSaveDirectory()` and `tests/TestSaveSandbox.hpp` ; every one of the 23 `verify_*` harnesses now points `Serializer` at a fresh temporary directory from `main()` , and every hand-rolled backup/restore dance in the test files was deleted outright rather than patched. A new regression case, `testTheSuiteCannotTouchRealSaveData` , asserts the sandbox's own containment; removing the sandbox to check that this case actually catches the regression reproduced the original bug exactly (it deleted the real `progress.json`). Result: `ctest` 13/13, and `verify_regressions.cpp` went from 497/498 (one failing, disk-state-dependent case) to 504/504.

Not verified / correction to the sweep plan

- The sweep plan (`docs/issues/submission_sweep_plan_2026-09-02.md` §4.3) states "Member B's share in W11–W13 is small (5 commits by `FubuGold` since 08-16)." That count is correct for the full span from 2026-08-16 to today (2026-09-02) — it includes 3 commits dated 2026-09-01 that fall inside W13, not W11 (`git log --all --author=FubuGold --since='2026-08-16'` → 5 commits total). Within the W11 window specifically (2026-08-16 through 2026-08-22), `FubuGold` has exactly **2** commits: `d954a66` (2026-08-16) and `5bc27c1` (2026-08-22). This report counts only those 2 for W11 and does not carry the plan's aggregate figure into this week's tally.
- The commit `5bc27c1` 's log entry claims to have "resolved" the Level 1-3 crash, but its diff contains no source changes — the actual fix is `4f80862` , authored earlier the same day by Member A. This report treats `5bc27c1` as a verification commit (see Issue 1), not as an independent fix, to avoid double-crediting the same defect.
- Whether the Bowser stagger fight and the P-Switch/POW effects introduced in `4f80862` play well (rather than merely run without crashing) is explicitly flagged as unplayed in the source log entry itself ("NOT PLAYED THROUGH... covered by ctest cases and by reading, not by observation") — this report repeats that flag rather than upgrading it to "verified."